# Evaluating Contextual Input and Shared Representation in Multi-Metric Runtime Forecasting for Cloud Software Systems

Abdelhak Kadouma[a,*], Amit Shukla[a] and Mohammed Elmusrati[a]

[a] University of Vaasa, Vaasa, Finland

ARTICLE INFO

*Keywords*:
runtime forecasting
cloud telemetry
multi-metric forecasting
multi-task learning
contextual input
shared representations

ABSTRACT

Multi-metric runtime forecasting requires decisions about other-resource telemetry and cross-target representation sharing, with consequences for accuracy and per-entity model consolidation. Our frozen experiment compares conventional long short-term memory (LSTM) formulations: S1 uses own-resource input and independent target models; S2 uses all-resource input and independent target models; M1 uses all-resource input and a shared multi-resource model. We evaluate central processing unit (CPU) utilisation, throughput, memory and disk-write activity across TimeTrack development data, Bitbrains fastStorage and Rnd external environments, and Materna T1–T3 dependent temporal blocks. Separate post-hoc controls constrain S2 to M1-selected contexts and increase M1's hidden size from H64 to H129 to approximately match S2's aggregate parameter count. A matched-context comparison of Ridge autoregression (AR) and autoregression with cross-resource inputs (ARX) examines contextual value beyond LSTMs. Contextual value is selective and model- and environment-dependent; memory benefits recur but are not universal. Adverse M1:S2 throughput, memory and disk-write point estimates persist across all six blocks under both controls, without identifying sharing alone as the cause. Yet M1 matched-cell median MAE ratios are close to one in many comparisons, and some larger MacroMAE penalties are tail-concentrated. Per-entity consolidation therefore presents a target-specific accuracy–consolidation trade-off. Ridge indicates model-family dependence. Effects can change sign across the recorded seeds, and sign stability need not imply magnitude stability. Relative gains do not ensure superiority to persistence. Contextual input and representation sharing require separate, resource-specific evaluation.

## 1. Introduction

Cloud telemetry forecasting can inform adaptation and resource management (Taherizadeh, Jones, Taylor, Zhao and Stankovski, 2018; Bauer, Züfle, Herbst, Zehe, Hotho and Kounev, 2020). Engineers must decide whether a target's forecaster should include other resources' telemetry and whether several targets should share a predictive representation. These choices change input information and model organisation, respectively. Consolidation reduces the models to manage but need not improve every resource's forecast. We evaluate accuracy and model consolidation, not downstream control effectiveness.

Established runtime-series benchmarking spans classical statistical, neural and zero-shot foundation models (Di Menna, Traini and Cortellessa, 2026). Cloud studies include multivariate resource inputs, recurrent architectures, joint multi-output prediction and hybrid models evaluated across multiple provider environments (Ullah, Bilal and Yoon, 2023; Tripathi, Priyadarshni, Misra and Singh, 2025). Global forecasting also learns parameters across collections of series (Salinas, Flunkert, Gasthaus and Januschowski, 2020). These precedents motivate empirical design choices for particular resource targets, not a new forecasting architecture.

*Corresponding author

abdelhak.kadouma@uwasa.fi (A. Kadouma); amit.shukla@uwasa.fi (A. Shukla); moel@uwasa.fi (M. Elmusrati)

*Contextual input* supplies other resources' histories to a target model without requiring shared parameters. *Representation sharing* learns several targets jointly. Multi-task learning offers inductive transfer through related tasks (Caruana, 1997), but task combinations and sharing arrangements require evaluation (Standley, Zamir, Chen, Guibas, Malik and Savarese, 2020; Fifty, Amid, Zhao, Yu, Anil and Finn, 2021; Nochumsohn, Zisling and Azencot, 2025). Independent all-input target models provide an intermediate comparison between own-resource and shared forecasting. MMRP already compares univariate/multivariate single-target CPU and memory forecasting (Tripathi et al., 2025); our bounded contribution is the within-study comparison including this intermediate organisation and joint resource forecasting (Section 2.4).

We evaluate conventional LSTM formulations with independent target models using own-resource histories (S1) or all-resource histories (S2), and a shared representation with resource-specific heads using all-resource histories (M1). S2 versus S1 evaluates contextual information, M1 versus S2 the implemented shared-LSTM transition, and M1 versus S1 the complete joint system. The evaluated resources are CPU utilisation, throughput, memory usage and disk-write activity. TimeTrack supplies development evidence, Bitbrains fastStorage and Rnd provide external environments, and Materna T1–T3 provide dependent temporal replication. The comparisons retain source-specific eligible populations and evaluation protocols. Sections 3 and 4 define the original formulations and evaluation.

The four questions in Section 4.1 address contextual input (RQ1), the operational shared-LSTM contrast (RQ2),

the complete joint system (RQ3), and heterogeneity and accuracy–efficiency trade-offs (RQ4).

Post-hoc sensitivity controls impose M1-selected contexts on S2, increase M1 from H64 to H129 to approximately match S2's aggregate capacity, and compare Ridge autoregression (AR) with cross-resource autoregression (ARX) at matched contexts. Section 4.11 details their separate selection procedures and limits; they qualify, rather than replace, the frozen results.

Contextual value is selective: memory shows recurring benefits, but directions depend on model family and environment. M1:S2 throughput, memory and disk-write point estimates remain adverse in all six blocks under both separate LSTM controls, without identifying sharing alone as the cause. Matched-cell median M1/S2 MAE ratios are close to one in many comparisons, whereas some much larger MacroMAE penalties are tail-concentrated (Section 5.2). Per-entity consolidation can therefore offer a target-specific trade-off, subject to assessment of adverse tails. Original external H64 M1 uses approximately 74.5% fewer parameters and 75% fewer models per target set. Seed sensitivity and poor persistence skill in some settings bound this accuracy–consolidation trade-off, which establishes no production savings (Section 6). We make four contributions:

1. We evaluate S1/S2/M1 operational contrasts that distinguish contextual-input value from the accuracy consequences of the implemented shared-LSTM design and its model consolidation.
2. We characterise resource-, environment- and model-dependent contextual value using the original LSTM comparison and a post-hoc matched-context Ridge AR/ARX comparison across all six evidence blocks.
3. We test the robustness of the shared-LSTM contrast with separate common-context and H129 approximate aggregate-capacity controls, distinguishing persistent adverse point estimates from causal claims about representation sharing alone.
4. We relate forecasting accuracy to model-management choices while bounding conclusions by matched-cell behaviour, existing-seed sensitivity, persistence skill and the eligible populations studied.

## 2. Related Work

### 2.1. Runtime and Cloud-Resource Forecasting

Runtime telemetry supports configuration and resource-management decisions in cloud and self-adaptive software systems (Taherizadeh et al., 2018; Bauer et al., 2020). Prediction-based provisioning and service-quality forecasting already connect forecast accuracy with software design (Calheiros, Masoumi, Ranjan and Buyya, 2015; Syu, Kuo and Fanjiang, 2017). Forecast quality nevertheless differs from controller effectiveness: policies, actuation delays and uncertainty also affect operational outcomes (Qu, Calheiros and Buyya, 2018). Here the engineering question concerns input selection and model management, not a measured autoscaling or service-level benefit.

The relevant literature spans different prediction organisations. Di Menna et al. (2026) compare model families on application runtime metrics, fitting conventional models independently per series and applying foundation models zero-shot to individual histories. In contrast, Ullah et al. (2023) use ten-channel histories and joint ten-channel forecasts with a common BiLSTM representation on fastStorage and Rnd. Multivariate covariates, hybrid resource predictors, joint outputs and multi-provider evaluation are established capabilities (Tran, Nguyen, Nguyen and Nguyen, 2018; Xu, Song, Wu, Gill, Ye and Xu, 2022; Tripathi et al., 2025; Rossi, Visentin, Carraro, Prestwich and Brown, 2025; Zhang, Yang, Yu and Pan, 2025), alongside online resource and microservice-indicator forecasting (Gao, Tang, Yao, Wu, Yuan, Zhou, Li and Chen, 2026b; Gao, Yu, Luo, Ma, Sun, Fang and Gao, 2026a). These precedents motivate comparing design choices, not claiming a new forecasting architecture.

TimeTrack provides OAI CI/CD telemetry with reported resource correlations (Meliani, Arora, Ksentini and Knopp, 2025). These motivate contextual inputs, but do not establish predictive value conditional on the target's own history. That value requires a forecasting comparison, not correlation alone.

### 2.2. Information Scope and Sharing in Forecasting

Input channels, output targets and parameter sharing are separate choices. A model may use several resource histories for one target; a vector output may represent several horizons of that same target rather than several resources. Independent target models require separate fitted parameters, not merely separate output values or separately processed channels.

MMRP directly compares univariate and multivariate single-target forecasts (Tripathi et al., 2025). Its CPU and memory results are reported separately; the exact all-channel assignment and independent-fit inventory are not fully specified. It is a substantive S1–S2-related precedent, not a reported joint resource-sharing arm. Likewise, esDNN uses resource covariates for CPU prediction (Xu et al., 2022). Neither multichannel input nor multi-head attention by itself establishes cross-target parameter sharing. Ullah's joint BiLSTM does share a representation, but its reported architecture comparisons do not include an independent all-input target-model bridge.

The distinction also matters outside cloud telemetry. PatchTST processes channels independently while reusing embedding and Transformer weights across them (Nie, Nguyen, Sinthong and Kalagnanam, 2023). Channel independence therefore does not mean independently fitted models. In contrast, iTransformer represents individual variates as tokens and models relationships between them through attention (Liu, Hu, Zhang, Wu, Wang, Ma and Long, 2024). DeepAR and local/global forecasting share functions across series rather than necessarily presenting all series as simultaneous covariates (Salinas et al., 2020; Montero-Manso and Hyndman, 2021). These distinctions matter when deciding

whether telemetry should be added to a predictor or target models should be consolidated.

### 2.3. Multi-Task Learning, Task Relatedness, and Adverse Transfer

Shared learning can provide an inductive bias (Caruana, 1997), but its value depends on tasks, data and sharing structure (Yu, Liu, Luo, Huang, Zhou, Liu, Hu, Chen, Zhang, Zhang, Shen, Adhikarla, Dai, Zhang, Kong, Ye, Yin, Namboodiri, Davison, Moore and Chen, 2025). Task grouping explicitly examines which tasks should learn together under accuracy and computational constraints (Standley et al., 2020; Fifty et al., 2021). Multi-objective and gradient-modification methods address conflicting updates (Sener and Koltun, 2018; Yu, Kumar, Gupta, Levine, Hausman and Finn, 2020). These are established responses to adverse transfer, not evidence of a measured mechanism in our resource forecasters.

Nikentari and Wei (2022) provide a forecasting precedent: NARMAX-LSTM combines lagged station inputs, shared LSTM layers and station-specific outputs, compares single-task baselines and discusses negative transfer. Its network-reduction argument is qualitative, and an independent all-input target-model bridge is not established. MTLinear assigns temporal weights to variate groups, analyses gradient conflicts and balance, and evaluates grouping, parameter size and computation (Nochumsohn et al., 2025). Applying common temporal weights to each variate's own history differs from mixing contextual channels. Its grouping endpoints inform independent-versus-shared learning without reproducing our all-input S2–M1 contrast.

Cloud multi-task forecasting also predates this study: Bi, Yuan, Zhou and Liu (2019) combine workload decomposition with stochastic configuration networks and report training-speed comparisons. The inspected evidence does not establish equivalence to resource-target sharing or an all-input independent baseline. Efficiency evidence must therefore be distinguished from a measured target-model consolidation trade-off. MMRP's inference-throughput measurements and Di Menna's model-size and adaptation-cost discussion likewise provide efficiency-related evidence, without the same target-consolidation analysis. Table 1 distinguishes these forms. Negative transfer and forecasting efficiency are not novel topics introduced here.

### 2.4. Evaluation Gap and Study Scope

The contribution concerns the within-study comparison, not an architecture or literature-wide priority claim. The inspected studies establish contextual single-target forecasting, shared forecasting, adverse-transfer analysis and efficiency evaluation in different designs. Table 1 records their overlap and evidence limits rather than certifying an absence across the literature.

S2 is the intermediate operational control: each target model receives all resource histories but retains separate fitted parameters. S1–S2 examines additional contextual input; S2–M1 examines the implemented move to a shared resource predictor. Capacity, optimisation and selected settings also differ, so these are not pure causal interventions or a complete $2 \times 2$ design. Separate post-hoc common-context and H129 capacity controls qualify that contrast, while matched-context Ridge AR/ARX checks contextual directions in another model family (Section 4.11).

The same operational contrasts are retained through TimeTrack development, the separate fastStorage and Rnd external environments, and dependent Materna T1–T3 temporal blocks. Resource-wise accuracy, local model consolidation and eligible-population limits define the evidence contribution. This supports runtime forecasting and model-management choices without claiming independent Materna datasets, a universal transfer mechanism or superiority to untested architectures. Section 3 defines the formulations.

## 3. Multi-Metric Runtime Forecasting Framework

The framework distinguishes access to resource histories from sharing a learned predictive representation across resource targets. Figure 1 summarises S1, S2 and M1 and their matched operational contrasts.

### 3.1. Problem Formulation

Let $\mathcal{R} = \{\text{cpu}, \text{thr}, \text{mem}, \text{disk}\}$ denote the resource set and let $R = |\mathcal{R}| = 4$. At discrete time $t$, the multivariate telemetry observation is $\mathbf{x}_t = (x_t^{(r)})_{r\in\mathcal{R}} \in \mathbb{R}^R$. For a context length $L \in \mathbb{N}$, the complete history available at forecasting origin $t$ is

$$\mathbf{X}_{t,L} = [\mathbf{x}_{t-L+1}, \dots, \mathbf{x}_t]^\top \in \mathbb{R}^{L\times R}. \tag{1}$$

Let $\mathcal{H} \subset \mathbb{N}_{>0}$ be a finite set of forecast horizons. For target resource $r$, $y_{t+h}^{(r)}$ is the future target at horizon $h \in \mathcal{H}$ and $\mathbf{y}_{t,\mathcal{H}}^{(r)} = (y_{t+h}^{(r)})_{h\in\mathcal{H}}$ is its multi-horizon target vector. A forecaster produces

$$\widehat{\mathbf{y}}_{t,\mathcal{H}}^{(r)} = F_r(\mathcal{I}_{t,L}; \boldsymbol{\theta}), \tag{2}$$

where $\mathcal{I}_{t,L}$ denotes the formulation-dependent information set and $\boldsymbol{\theta}$ denotes the relevant model parameters.

Models are entity-specific: for each TimeTrack machine or external VM, outer fold and seed, S1/S2 fit four target-specific networks and M1 one joint four-resource network. M1 shares across that entity's targets, never across entities. These local model-complement counts exclude the separate candidate-context inner fits and final refits; they do not describe a globally deployed four-versus-one system.

Contexts contain aligned observations available at or before $t$; targets lie strictly afterwards. Horizons use each dataset's time grid, preserving temporal order without requiring a universal sampling interval. Eligible origins supply every formulation's inputs and complete targets. Matched origins, targets, horizons and partitions prevent sample differences from changing a contrast.

**Table 1**
Closest inspected designs: contextual inputs, target sharing and model-management evidence. Codes describe reported designs under the definitions below, not architectural equivalence or a priority ranking.

| Study | Contextual input / target organisation | Independent all-input targets | Shared cross-target parameters | Single-task vs shared comparison | Adverse-transfer analysis | Efficiency / consolidation evidence |
|---|---|---|---|---|---|---|
| Ullah et al. (2023) | YES; ten-channel histories to joint ten-channel forecasts. | NO | YES; common BiLSTM | NO | NO | PARTIAL; qualitative training discussion |
| Di Menna et al. (2026) | NO; individual metric histories to same-metric horizons. | NO | NO; pretrained reuse is a different axis | NO | NO | PARTIAL; sizes and cost discussion |
| Tripathi et al. (2025) (MMRP) | YES; own/multichannel histories to one target. | PARTIAL; exact channel/fit inventory unspecified | NO | NO | NO | PARTIAL; inference throughput |
| Nikentari and Wei (2022) (NARMAX-LSTM) | YES; lagged station inputs to station targets. | Not established | YES; shared LSTM layers | YES; single-task baselines | YES; task-wise discussion | PARTIAL; network-reduction argument |
| Nochumsohn et al. (2025) (MTLinear) | NO channel mixing in the linear predictor; own histories, grouped targets. | NO | YES; weights within groups | PARTIAL; grouping endpoints | YES; gradient analysis | YES; parameter/grouping and compute analysis |
| Bi et al. (2019) | Not established for cross-resource inputs; decomposed workload. | Not established | Not established for resource targets | Not established | Not established | PARTIAL; training-speed comparison |
| This study | YES; S1 own history; S2/M1 all histories; independent/joint resource targets. | YES; S2 | YES; M1 encoder | YES; S2–M1 | YES; operational accuracy contrast | YES; model and parameter counts; worker fit-hours |

Contextual input means other variables enter a target's predictor, not merely that a dataset is multivariate. Independent all-input targets means separately fitted target models with the selected multichannel inputs. Sharing concerns learning across target series, not horizons alone or pretrained reuse. YES: explicit evidence; NO: excluded by the inspected reported design; PARTIAL: related, limited or incompletely matched evidence. Not established: available evidence does not settle the cell. Single-task/shared comparisons need not hold inputs or architecture fixed; adverse-transfer analysis coded YES means explicit empirical analysis of adverse-transfer outcomes, not causal identification of the mechanism. Efficiency PARTIAL covers timing, qualitative cost or network-reduction evidence without the quantified parameter/grouping or consolidation analysis coded YES. No code rules out unpublished implementations.

### 3.2. Resource Metrics and Multi-Horizon Forecasting

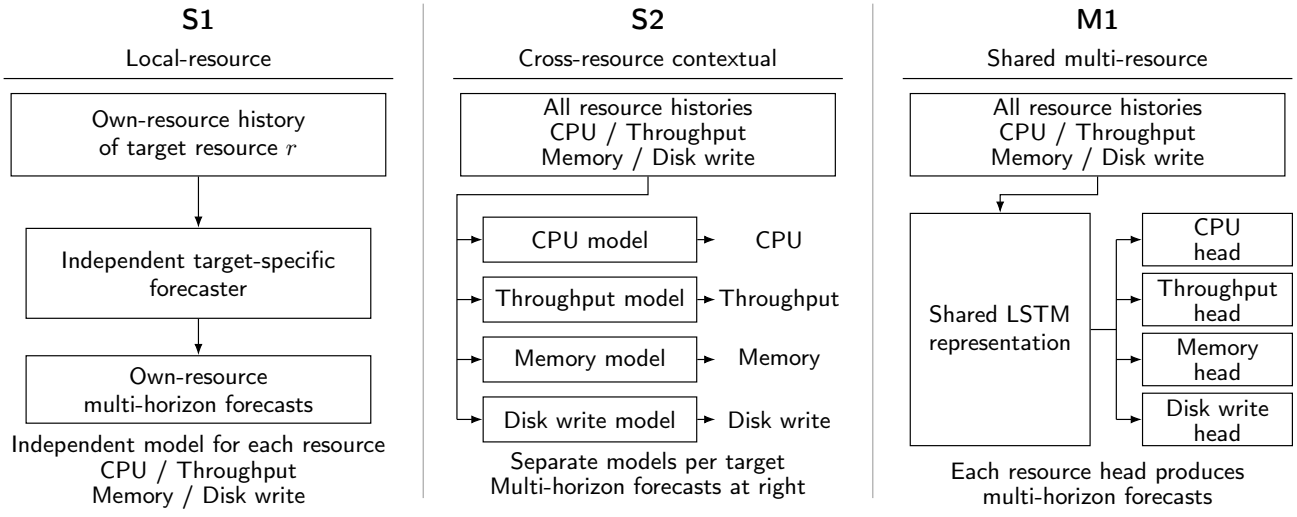
The resource vector comprises CPU utilisation, aggregate throughput, memory usage, and disk-write activity. CPU utilisation represents processor demand; throughput represents the aggregate rate of data transferred through the runtime environment; memory usage represents occupied memory; and disk-write activity represents write traffic to persistent storage. Dataset-specific fields, units and sampling intervals are mapped in Section 4.2.

Each formulation jointly predicts its full horizon vector, so output organisation is fixed rather than changed from single- to multi-horizon training. Contexts $L$ and finite positive offsets $\mathcal{H}$ are fixed before evaluation but may vary with source cadence without changing the formulation definitions. Distinct resources may supply predictive context; varying access to their histories tests this while keeping targets fixed. Resource-specific training transforms and scaling accommodate different distributions and optimisation scales; evaluation returns forecasts to native units (Section 4.3).

### 3.3. S1 Local-Resource Forecasting

S1 is the local-information formulation. For target resource $r$, its input is the single-channel sequence

$$\mathbf{X}_{t,L}^{\mathrm{S1},r} = [x_{t-L+1}^{(r)}, \dots, x_t^{(r)}]^\top \in \mathbb{R}^{L\times 1}, \tag{3}$$




| | |
|---|---|
| **S2 versus S1** | Cross-resource contextual information |
| **M1 versus S2** | Implemented representation sharing |
| **M1 versus S1** | Complete joint-system change |

**Figure 1:** S1 fits independent target models from own-resource histories; S2 supplies all histories to each independently trained target model. M1 uses all histories with a shared LSTM representation and resource-specific heads. S2 versus S1 assesses contextual input; M1 versus S2 the implemented move to representation sharing; M1 versus S1 the complete joint-system change. Models are fitted separately for each entity, outer fold and seed; S1/S2 use four networks per target set and M1 uses one joint network. These are matched operational comparisons, not a complete 2×2 design, parameter-matched interventions or pure causal isolation of either design choice.

and its output is

$$\widehat{\mathbf{y}}_{t,\mathcal{H}}^{(r)} = f_r^{\mathrm{S1}}(\mathbf{X}_{t,L}^{\mathrm{S1},r}; \boldsymbol{\theta}_r^{\mathrm{S1}}) \in \mathbb{R}^{|\mathcal{H}|}. \tag{4}$$

A distinct parameter set $\boldsymbol{\theta}_r^{\mathrm{S1}}$ gives each resource its own encoder, prediction mapping and objective. S1 is the multi-horizon local-history baseline within the common model family; withholding non-target channels does not assume resource independence.

### 3.4. S2 Cross-Resource Contextual Forecasting

S2 expands the information set while retaining target-specific modelling. For every $r \in \mathcal{R}$, the input is the complete history

$$\mathbf{X}_{t,L}^{\mathrm{S2},r} = \mathbf{X}_{t,L} \in \mathbb{R}^{L\times R}, \qquad \widehat{\mathbf{y}}_{t,\mathcal{H}}^{(r)} = f_r^{\mathrm{S2}}(\mathbf{X}_{t,L}; \boldsymbol{\theta}_r^{\mathrm{S2}}). \tag{5}$$

Each $\boldsymbol{\theta}_r^{\mathrm{S2}}$ is trained independently. Non-target histories act as covariates, but the output and loss remain specific to $r$: gradients from one target cannot update another model's representation. Thus all-resource input does not establish shared learning across targets.

Under matched support, S2 versus S1 evaluates contextual input while retaining separate models, without presuming that extra channels are informative or that their contribution has a particular sign.

### 3.5. M1 Shared Multi-Resource Forecasting

M1 combines all-resource input with a representation shared across target tasks. A temporal encoder $E_{\boldsymbol{\theta}}$ maps the complete context to a latent vector $\mathbf{z}_t$, and one resource-specific head $g_{r,\boldsymbol{\phi}_r}$ maps that representation to the horizon vector for resource $r$:

$$\begin{aligned} \mathbf{z}_t &= E_{\boldsymbol{\theta}}(\mathbf{X}_{t,L}), \\ \widehat{\mathbf{y}}_{t,\mathcal{H}}^{(r)} &= g_{r,\boldsymbol{\phi}_r}(\mathbf{z}_t), \quad r \in \mathcal{R}. \end{aligned} \tag{6}$$

The resulting output has resource-by-horizon structure $\widehat{\mathbf{Y}}_t \in \mathbb{R}^{R\times|\mathcal{H}|}$. The common encoder is updated by all resource losses; head gradients update only the corresponding head, retaining resource-specific predictive mappings. This single multi-output model introduces no attention, gating, task routing or adaptive sharing.

Let $\mathcal{B}$ be a training mini-batch, and let $\widetilde{y}_{i,h}^{(r)}$ and $\widetilde{\widehat{y}}_{i,h}^{(r)}$ denote the scaled target and prediction for sample $i$, resource $r$, and horizon $h$. M1 uses the equally weighted scaled-space objective

$$\mathcal{L}_{\mathrm{M1}} = \frac{1}{|\mathcal{B}|R|\mathcal{H}|} \sum_{i\in\mathcal{B}} \sum_{r\in\mathcal{R}} \sum_{h\in\mathcal{H}} \left( \widetilde{\widehat{y}}_{i,h}^{(r)} - \widetilde{y}_{i,h}^{(r)} \right)^2. \tag{7}$$

This is the arithmetic mean of four transformed/robust-scaled resource MSEs, with equally weighted horizons. S1/S2 use the corresponding single-resource MSE. Equal coefficients do not imply equal realised losses or gradients: residual distributions and derivatives differ. No task weights are learned and no resource is conditionally suppressed. Scaling is training-only (Section 4.3).

### 3.6. Controlled Information-Sharing Decomposition

The three matched operational formulations define three comparisons, each evaluated for the same resource, horizon, and matched support. Let $Q(F)$ denote a lower-is-better error functional for formulation $F$ under such a fixed evaluation unit. For a baseline $A$ and candidate $B$, define the signed error reduction

$$D_Q(A \to B) = Q(A) - Q(B), \tag{8}$$

so that a positive value indicates lower error for $B$. Exact metric, aggregation, and percentage-normalisation rules are specified in the experimental methodology. The framework-level contrasts are

$$\begin{aligned} \mathcal{C}_{\text{context}} &= D_Q(\text{S1} \to \text{S2}), \\ \mathcal{C}_{\text{sharing}} &= D_Q(\text{S2} \to \text{M1}), \\ \mathcal{C}_{\text{joint}} &= D_Q(\text{S1} \to \text{M1}). \end{aligned} \tag{9}$$

The context contrast retains separate target models; the sharing contrast retains all-resource inputs; the joint contrast combines both changes. None presumes an error ordering. Comparisons match eligible origins, targets, resource/horizon coordinates and partitions, and never average heterogeneous native-unit resource errors. On this support the S1-to-M1 raw difference in $Q$ equals the sum of its two intermediate differences. This algebra does not imply mechanistic additivity or causal isolation. Reported percentages are not generally additive because each uses its own reference error.

## 4. Experimental Methodology

The formulations in Section 3 are retained across one development dataset, two external environments and three dependent temporal blocks, adapting source-specific fields and sample offsets.

### 4.1. Research Questions

Four research questions use the original matched contrasts in Section 3.6.

**RQ1:** Does cross-resource context improve forecasting relative to local-resource forecasting? This question is evaluated by comparing S2 with S1, which changes the input from the target resource alone to all four resource histories while retaining separate target-specific models.

**RQ2:** How does the implemented shared-LSTM formulation compare with separate contextual target models? This question is evaluated by comparing M1 with S2, for which both formulations receive all-resource input but only M1 shares a temporal encoder across target tasks.

**RQ3:** Does the complete shared multi-metric system outperform local-resource forecasting? This end-to-end question compares M1 with S1 and therefore measures the joint change in input scope and representation sharing.

**RQ4:** How stable are these effects across resources, environments, temporal blocks, horizons, and evaluation units, and what accuracy–efficiency trade-offs arise? Stability is examined without collapsing the resource-specific contrasts into heterogeneous native-unit errors.

These specify estimands, not expected benefits; efficiency is assessed separately from accuracy.

### 4.2. Datasets

Table 2 summarises the experimental environments and their roles. All sources are mapped to the same four resource semantics, but their sampling grids, eligible populations, and evidential roles remain distinct.

#### *4.2.1. TimeTrack Telemetry*

TimeTrack is the framework-development source. It contains telemetry from the OpenAirInterface (OAI) CI/CD system. Seven machines were observed between 24 June and 19 July 2024. The canonical panel is the exact sorted intersection of timestamps across the six source files; no nearest-time matching, resampling, interpolation, imputation, or value cleaning is used. Its measured median sampling interval is 42.285164 seconds.

Discontinuities are represented explicitly. A new canonical segment begins when adjacent retained timestamps are separated by more than 84.57 seconds, twice the median interval, yielding nine segments. Historical windows and future targets cannot cross these boundaries. The four study targets are machine-level CPU utilisation, aggregate bidirectional throughput, used memory, and disk-write activity. Throughput is constructed for each machine as the sum of validated `bond0` transmit and receive rates; used-memory and disk-write fields are retained at machine level. TimeTrack supplies internal development evidence and is not treated as an independent external dataset.

#### *4.2.2. Bitbrains fastStorage and Rnd*

The Bitbrains fastStorage and Rnd traces are kept as two separate external confirmation environments. CPU, network receive plus transmit throughput, memory usage, and disk-write throughput are mapped respectively to the four common resource targets. Rows that are physically invalid are removed without repair, duplicate timestamps are rejected, and gaps longer than 450 seconds delimit contiguous segments.

Table 2 distinguishes the structural and scale-eligible pools. A prospective training-scale amendment required positive transformed-resource interquartile range in every required training partition. From each restricted pool, VMs were selected by ascending SHA-256 rank of a frozen environment-and-identifier key. The rule is deterministic and uses neither predictive outcomes nor manual substitution.

**Table 2**
Evidence sources, study roles and frozen protocol adaptations, distinguishing eligible populations from executed cohorts. Materna T1–T3 are dependent temporal blocks.

| Source / block | Study role | Temporal structure / cadence | Eligibility | Executed cohort | Forecast protocol |
|---|---|---|---|---|---|
| TimeTrack | Framework development | 42.285 s median; 9 gap-bounded segments | Seven machines; 41,356-row canonical panel | 7 machines | 0.705–90.208 min (eight leads) |
| Bitbrains fastStorage | External confirmation | Separate environment; 5 min; per-VM segments | 916/1,250 structural; 596/916 scale eligible | 35 VMs | 5, 10, 20, 45, 90 min (five leads) |
| Bitbrains Rnd | External confirmation | Separate environment; 5 min; per-VM segments | 416/500 structural; 272/416 scale eligible | 35 VMs | 5, 10, 20, 45, 90 min (five leads) |
| Materna T1–T3 | Secondary replication | 5 min; blocks analysed separately | 398 common structural; 74/398 all-block scale | 35 shared VMs | 5, 10, 20, 45, 90 min (five leads) |

All experiments forecast CPU utilisation, throughput, memory usage and disk-write activity. Materna provides secondary temporal replication: T1–T3 are consecutive dependent blocks from the same source and selected cohort, not independent datasets.

The selected cohorts are therefore samples from scale-estimable, structurally eligible populations; they are not claimed to represent every Bitbrains VM.

#### *4.2.3. Materna Temporal Traces*

Materna provides a secondary temporal replication. T1, T2, and T3 are consecutive blocks from the same source, not independent datasets. They are processed independently on their native 300-second grid, with contiguous segments defined per trace and VM. The resource mapping is CPU usage percentage, received plus transmitted network throughput in KB/s, memory usage in KB, and disk-write throughput in KB/s.

Applying the strict training-scale rule jointly across every trace, fold, training phase, context, and resource retains 74 VMs, or 18.5930% of the common structural pool. The first 35 VMs under the frozen ascending SHA-256 rank are used in all three blocks, preserving the same identifiers wherever structurally valid and permitting matched temporal comparisons. T1–T3 are analysed first as separate temporal blocks. Any combined estimate uses equal block weights; the blocks are never treated as independent observations in a meta-analysis.

### 4.3. Target Construction and Preprocessing

Dataset-specific fields and units are mapped to the semantic order CPU, throughput, memory, and disk write. CPU and memory use the identity transform. Throughput is the sum of receive and transmit rates, or the frozen equivalent aggregate in TimeTrack, and uses $\log(1 + x)$. Disk-write activity or throughput uses the same $\log(1 + x)$ transform. These definitions preserve the meaning of each resource while accommodating different source schemas and native units.

At each legal forecasting origin, these mappings yield a four-resource history and future resource-by-horizon target tuple. Direct multi-horizon prediction is used: every head estimates all prescribed future offsets in one forward pass, rather than recursively feeding an earlier prediction into a later horizon. The target construction is identical across formulations on a given dataset. Thus S1 receives only the target channel's history, S2 receives all four histories for the same single target, and M1 receives all four histories and all four target vectors. Missing or invalid values remove the affected origin from common support; they are not repaired differently for one formulation.

For resource $r$, let $g_r$ denote its fixed transform. Within a legal training partition, transformed values are robustly scaled as

$$\begin{aligned} \widetilde{y}^{(r)} &= \frac{g_r(y^{(r)}) - m_r}{q_r}, \\ m_r &= \operatorname{median}_{\text{train}}\, g_r(y^{(r)}), \\ q_r &= \operatorname{IQR}_{\text{train}}\, g_r(y^{(r)}). \end{aligned} \tag{10}$$

The interquartile range must satisfy $q_r > 0$. The final protocol uses no epsilon, fallback scale, or outcome-dependent exception. Inner scalers are fit only on inner-training data; after model selection, final scalers are refit on the complete legal outer-training partition. Validation and test values never contribute to scaler parameters.

CPU and memory predictions are returned to native units by inverse robust scaling. Throughput and disk-write predictions are inverse-scaled, passed through expm1, and deterministically projected onto the non-negative domain. The same transform, scaler, and inverse mapping are used for matched S1, S2, and M1 fits. TimeTrack uses eight output horizons whereas the external datasets use five; this changes output width but not the resource-transform semantics.

### 4.4. Temporal Support and Leakage Control

Forecasting origins are ordered chronologically within the support universe. An origin is legal only when the maximum candidate context and every required future target are finite and remain in the same contiguous temporal segment. Common support is constructed at the largest context, so shorter contexts gain no additional origins. S1, S2, and M1 consequently use identical origin IDs, target IDs, machines or VMs, folds, horizons, and seeds within every comparison.

Five expanding outer folds are used for each machine, VM, environment, or trace. The first half of the ordered legal origins precedes the first test block; five successive, non-overlapping ten-percent blocks form the outer tests. Earlier test periods may enter the training history of a later fold. For each fold, an otherwise eligible training origin is removed unless its maximum-horizon target timestamp is strictly earlier than the first outer-test origin. This target-time purge prevents a training label from extending into the test period even when its forecasting origin precedes the boundary.

Within the purged outer-training support, the chronologically latest 15% of origins is provisional inner validation. Earlier inner-training origins are purged again against the first validation origin using the maximum forecast horizon. There is no shuffling. Scalers, model parameters, context selection, and epoch selection can use only the relevant legal training and validation partitions; outer-test observations cannot determine filtering, scaling, context, epoch count, or fitted parameters.

After context and seed-specific epoch selection, each final model is reinitialised and refit once on the complete legal outer-training support with the selected fixed epoch count and a newly fitted outer-training scaler. It has no validation or test callback. Each successful final fit is then evaluated exactly once on its outer-test block. These controls separate model selection from evaluation and ensure that formulation effects are not attributable to different temporal support.

### 4.5. Forecast Horizons and Contexts

TimeTrack uses sample leads $\{1, 2, 4, 8, 16, 32, 64, 128\}$. At its measured median interval, these correspond approximately to 0.705, 1.410, 2.819, 5.638, 11.276, 22.552, 45.104, and 90.208 minutes. Candidate context lengths are 32, 64, and 128 samples, covering approximately 22.6, 45.1, and 90.2 minutes.

Bitbrains and Materna use five-minute sampling, so their sample leads are $\{1, 2, 4, 9, 18\}$, corresponding exactly to 5, 10, 20, 45, and 90 minutes. Their candidate contexts are 4, 9, and 18 samples, or approximately 20, 45, and 90 minutes. The integer offsets therefore differ by dataset. The adaptation preserves a comparable real-time forecast and context range rather than pretending that one sample index has the same elapsed-time meaning on all grids.

### 4.6. Model Architecture and Training

In the original frozen experiment, all formulations use a plain LSTM (Hochreiter and Schmidhuber, 1997): a two-layer unidirectional LSTM with hidden size 64, dropout 0.1, and the final recurrent output as its temporal representation. S1 has one input channel and one resource-specific horizon head; S2 has four input channels but remains one independent model per target; M1 has four inputs, one shared encoder, and four independent resource heads. S1 and S2 minimise target-specific scaled mean squared error, whereas M1 minimises the arithmetic mean of the four resource-specific scaled mean squared errors. Inner early stopping minimises validation MSE in this transformed/robust-scaled space and records a best epoch separately for each seed.

Training uses Adam with learning rate 0.001, batch size 256, at most 50 epochs, early-stopping patience 7 during inner fitting, gradient-norm clipping at 1.0, and seeds 0, 1, and 2. No architecture or optimiser search is performed. The plain backbone is deliberately simple because the experiment compares the operational formulations rather than proposing an architecture-specific innovation.

Output width affects parameter counts. For the five-horizon external design, S1 contains 50,757 parameters per resource model (203,028 across four), S2 contains 51,525 per target model (206,100 across four), and M1 contains 52,500 in its single shared model. For eight-horizon TimeTrack, the corresponding counts are 50,952 per S1 model (203,808 across four), 51,720 per S2 model (206,880 across four), and 53,280 for M1. These H64 counts refer to one entity/fold/seed target complement, not parameters pooled across the corpus. Model-complement and total training-fit counts differ as defined in Section 3.1; original parameter budgets are unequal.

Reported fitting time is the sum of successful unique worker fit-hours across inner candidate fits and final refits. Inner durations include training-loop and validation-forward time; final durations cover the training loop. Unsuccessful attempts, skipped/resumed encounters, model and window construction, preprocessing, checkpoint serialisation, final prediction and inversion, and orchestration are excluded. Summed worker fit-hours are not elapsed wall time, energy, deployment cost or production latency. Differences between S1 and S2 can reflect executed epochs and selected context lengths; input dimensionality alone does not explain their timing order.

### 4.7. Context and Model Selection

In the original LSTM experiment, context length is selected from the frozen candidate grid while architecture and optimiser settings remain fixed. For S1 and S2, one context is selected independently for each environment or trace, machine or VM, outer fold, and target resource. The selection score is validation MAE of the transformed/robust-scaled targets and predictions, evaluated at each seed's MSE-selected best epoch, then averaged equally over horizons and the three seeds. M1 selects one context per environment or trace, machine or VM, and fold by taking the equal mean of the same scaled-space validation MAE over the four resources, all horizons and all seeds. Exact ties are resolved in favour of the shorter context.

Early stopping and context selection consequently use different error functionals in the same training space: scaled MSE chooses the epoch, and scaled MAE scores the context at that epoch. This context score is not native evaluation NMAE. For log-transformed targets, inverse transformation and non-negativity projection need not preserve candidate rankings. The selected context, same seed and seed-specific fixed epoch determine the fresh final refit in Section 4.4;

inner weights are not inherited. Selection precedes outer-test evaluation. Target-specific S1/S2 contexts and M1's common context reflect their separate-model and shared-encoder organisations, not broad architecture or hyperparameter tuning.

### 4.8. Evaluation Metrics and Statistical Analysis

Let $y_i$, $\hat{y}_i$, and $\bar{y}$ denote the observed value, forecast, and observed mean for $n$ matched cases of one resource and horizon. Evaluation uses

$$\begin{aligned} \mathrm{MAE} &= \frac{1}{n}\sum_{i=1}^{n}|y_i-\hat{y}_i|, \\ \mathrm{RMSE} &= \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}, \\ R^2 &= 1-\frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2}, \\ \mathrm{NMAE}_r &= \frac{\mathrm{MAE}_r}{q_r^{\mathrm{native}}}, \end{aligned} \tag{11}$$

where $q_r^{\mathrm{native}}$ is the native-unit outer-training IQR for resource $r$, distinct from the transformed training IQR $q_r$ in Equation 10. Primary evaluation follows inverse transformation and the prescribed non-negativity projection. MAE and RMSE are reported in each resource's native units; $R^2$ and NMAE are unitless. Native errors are never averaged across different resources.

Persistence supplies a separate absolute forecasting sanity check:

$$\hat{y}_{t+h}^{(r),\mathrm{pers}} = y_t^{(r)}, \qquad \mathrm{Skill}_{r,h} = 1-\frac{\mathrm{MAE}_{r,h}^{\mathrm{model}}}{\mathrm{MAE}_{r,h}^{\mathrm{pers}}}. \tag{12}$$

Persistence skill does not replace the controlled formulation comparisons: relative improvement over another learned formulation does not imply absolute superiority over persistence. Reported overall skill averages the existing per-cell relative skills over entity–fold cells and then horizons; it is not $1-\mathrm{MacroMAE}_{\mathrm{model}}\,/\,\mathrm{MacroMAE}_{\mathrm{pers}}$. Undefined zero-denominator skills are excluded from this ancillary mean with their counts retained, not replaced by zero.

The primary aggregation begins by averaging stochastic-model MAE across the three seeds within each matched environment or trace, entity, fold, resource, horizon, and formulation cell. For a fixed resource and horizon, those seed-averaged entity–fold cells receive equal weight to form MacroMAE. A resource-overall MacroMAE then gives equal weight to every horizon. For candidate $C$ and baseline $B$, the reported gain is

$$G(C:B) = 100\,\frac{\mathrm{MacroMAE}(B)-\mathrm{MacroMAE}(C)}{\mathrm{MacroMAE}(B)}. \tag{13}$$

A positive value indicates lower candidate error. The three contrasts are S2 versus S1, M1 versus S2, and M1 versus S1. Secondary diagnostics include pooled prediction-level gain, mean and median matched-cell relative gain, and the fractions or counts of positive VMs, folds, horizons, and seed aggregates.

Prospectively frozen directional classifications combine overall gain magnitude with horizon, entity, fold and seed consistency. A positive aggregate has gain strictly greater than zero; exact zero is not positive. Entity signs aggregate over folds and horizons, fold signs over entities and horizons, and seed signs over entities, folds and horizons. The following conditions are conjunctive, with overall gain expressed in percent:

**Broad Positive.** Overall gain is at least $+1\%$, with at least $6/8$ horizons, $5/7$ machines, $4/5$ folds and $2/3$ seeds positive in TimeTrack. Externally, the horizon and entity thresholds are $4/5$ and $25/35$ VMs; fold and seed thresholds are unchanged.

**Conditionally Positive.** The broad-positive rule fails, overall gain is strictly positive, and at least $4/8$ horizons, $3/7$ machines, $3/5$ folds and $2/3$ seeds are positive in TimeTrack. Externally, the horizon and entity thresholds are $3/5$ and $15/35$ VMs.

**Broad Negative.** Overall gain is at most $-1\%$, with at most $2/8$ horizons, $2/7$ machines, $1/5$ folds and $1/3$ seeds positive in TimeTrack. Externally, the horizon and entity limits are $1/5$ and $10/35$ VMs.

**Neutral or Mixed.** Every remaining case, including point gains whose magnitude or lower-level consistency fails these rules.

These classes are not labels assigned solely by the sign of the primary point estimate. Bootstrap intervals quantify a separate form of uncertainty: neither interval exclusion of zero nor a significance test enters the classification predicates. Full trace-level and RQ1–RQ4 predicates, including precedence and temporal classification-change rules, remain in the supplementary decision criteria and frozen protocol. No threshold is adapted after observing outcomes. For external uncertainty, a paired VM-cluster bootstrap samples 35 VMs with replacement, retains all five folds and the paired formulation structure, and operates on seed-averaged matched MAEs. It uses 10,000 replicates, percentile 95% intervals (2.5th and 97.5th percentiles), and random seed 20261007. The bootstrap is performed independently within each Bitbrains environment and each Materna trace. Materna blocks are not resampled as independent datasets, and this exact external bootstrap is not retrospectively imposed on earlier exploratory TimeTrack analyses.

### 4.9. External and Temporal Replication Design

The source roles in Section 4.2 retain the formulations, resource semantics, matched support and comparable real-time forecast range, while allowing source-specific fields,

units and offsets. No score pools all roles. Bitbrains environments are analysed before their unweighted equal-environment summary; Materna traces precede their descriptive synthesis, for a fixed comparison and resource:

$$G_{\text{temporal}} = \frac{G_{\text{T1}} + G_{\text{T2}} + G_{\text{T3}}}{3}. \tag{14}$$

Equal temporal-block weights replace weighting by rows, origins, predictions, duration or fit count. Cross-source synthesis describes directions and frozen class consistency, not a pooled error, global interval or independent-dataset meta-analysis.

### 4.10. Population Restrictions

External eligibility requires both structural validity and non-degenerate training variability. Specifically, every transformed CPU, throughput, memory, and disk-write series must have IQR strictly greater than zero in every required scaler-fit partition. The partitions cover each fold, inner and final training phase, and candidate context; for Materna the same requirement also holds in every temporal block. There is no epsilon, fallback, resource mask, or formulation-specific exemption.

Table 2 reports structural, scale-eligible and executed populations. Deterministic hash ranking follows eligibility, not random or performance-ranked selection. The estimand concerns eligible dynamic, scale-estimable multi-resource VMs; reproducibility does not make the cohort a probability sample of all archived VMs. Excluded idle, quiescent, constant, intermittently observed or structurally invalid workloads remain outside scope.

### 4.11. Post-Hoc Sensitivity Analyses

Three completed analyses address reviewer challenges after the original experiment was frozen. They are explicitly *post-hoc sensitivity controls*, not amendments to the original prospective design or replacements for its primary results. All retain the frozen cohorts, outer folds, forecast horizons, purging, legal support, source-specific transforms, training-only scalers and native-unit evaluation. Models remain entity-specific, with no cross-machine or cross-VM pooling. Results are reported separately by block and resource.

*S2 with common context.* For each entity and outer fold, the frozen original H64-M1 context is imposed on every independent S2 target. Each seed retains the S2 MSE-selected epoch from its existing inner fit at that imposed context. Exact matching final artifacts are reused only after identity, support, scaler and hash checks; otherwise the selected fixed-epoch outer-training refit is performed. This removes S2's target-specific context-selection freedom relative to H64 M1, but does not equalise parameter count, architecture or optimisation.

*M1 with approximately matched aggregate capacity.* The shared LSTM hidden size increases from H64 to H129, giving 206,420 parameters externally and 207,980 for TimeTrack, approximately matching the corresponding four-network S2 complements of 206,100 and 206,880. All other scientific settings are retained. M1's original inner context and seed-specific epoch selection are repeated afresh, followed by fresh final refits; H64 choices and fitted weights are not reused. This controls the approximate aggregate parameter budget, not task-specific capacity, architecture or optimisation equivalence. Common-context S2 remains aligned to H64-M1 contexts, not necessarily the newly selected H129 contexts; the two LSTM controls are separate, not a jointly matched intervention.

*Matched-context Ridge AR/ARX.* AR uses only the target resource's lag window; ARX uses all four resource lag windows. Both use the same frozen H64-M1 context per entity and fold, without context tuning, and directly predict the complete horizon vector for one target. The existing multi-output Ridge implementation flattens each window and applies an additional lag-feature StandardScaler fitted only on training data. Targets retain the source-specific transformed/robust-scaled space. For each formulation and target, $\alpha \in \{0.01, 0.1, 1, 10, 100\}$ is selected strictly within the inner split using native-unit validation MAE after inverse transformation/projection, averaged equally over horizons; exact ties favour the smaller alpha. The selected model is refitted on outer-training data and evaluated once on outer test. Ridge is deterministic, so no artificial seed repetitions are introduced. This tests input-scope sensitivity within a second model family, not equality of LSTM/Ridge selection objectives or a causal isolation of model family from the specified context and model design. These post-hoc analyses introduce no additional significance test or pooled cross-block performance statistic.

## 5. Results

Figure 2 reports frozen resource-overall MacroMAE gains for each evidence block: positive favours the candidate, negative the baseline. Section 4.8 distinguishes these estimates from frozen consistency classes and prospective VM-cluster intervals; TimeTrack has no such external-bootstrap intervals. Block comparisons remain descriptive under the source roles in Section 4.9. The *POST-HOC* controls and existing-artifact diagnostics supplement, not replace, primary estimates, classes or intervals. Table 3 uses the designs in Section 4.11 beside unchanged reference gains. Supplementary Tables S1–S2 provide the detailed diagnostics.

### 5.1. RQ1 Cross-Resource Context

RQ1 compares independent all-resource S2 models with own-resource S1 models. Figure 2(a) shows selective contextual value, not uniform improvement. In TimeTrack, memory was the sole positive resource-level effect, while all four frozen classifications remained neutral or mixed because the required consistency thresholds were not met.

The Bitbrains environments did not reproduce one common direction. In fastStorage, memory was broadly positive

**(a) Contextual input: S2 versus S1**

| | TimeTrack | fastStorage | Rnd | Materna temporal blocks T1 | T2 | T3 |
|---|---|---|---|---|---|---|
| CPU | -3.31 | -3.20 | +1.41 | -4.32 | +4.21 | -2.79 |
| Throughput | -0.24 | +0.01 | -7.21 | +0.99 | +0.29 | +0.04 |
| Memory | +0.86 | +2.45 | +1.67 | -0.02 | +3.95 | +3.09 |
| Disk write | -0.49 | +2.69 | +0.16 | -6.70 | -1.12 | +1.91 |

**(b) Shared representation: M1 versus S2**

| | TimeTrack | fastStorage | Rnd | Materna temporal blocks T1 | T2 | T3 |
|---|---|---|---|---|---|---|
| CPU | +0.47 | +0.05 | -16.51 | +0.88 | -2.56 | -8.10 |
| Throughput | -0.84 | -27.09 | -14.32 | -7.31 | -8.69 | -7.91 |
| Memory | -18.48 | -4.87 | -8.92 | -13.22 | -9.27 | -8.70 |
| Disk write | -0.80 | -4.07 | -5.55 | -27.46 | -7.89 | -71.30 |

**(c) Full joint system: M1 versus S1**

| | TimeTrack | fastStorage | Rnd | Materna temporal blocks T1 | T2 | T3 |
|---|---|---|---|---|---|---|
| CPU | -2.83 | -3.15 | -14.86 | -3.40 | +1.75 | -11.11 |
| Throughput | -1.08 | -27.08 | -22.57 | -6.25 | -8.37 | -7.87 |
| Memory | -17.46 | -2.30 | -7.09 | -13.24 | -4.95 | -5.34 |
| Disk write | -1.29 | -1.27 | -5.38 | -36.00 | -9.10 | -68.03 |

Nonlinear asinh colour aid; exact values printed above

Gain (%)

-75 -20 -5 -1 0 +1 +5 +20 +75

**Figure 2:** Primary overall MacroMAE percentage gains for (a) S2 versus S1, (b) M1 versus S2 and (c) M1 versus S1 across the evaluated evidence blocks. Positive values indicate lower MacroMAE for the candidate formulation; negative values indicate higher MacroMAE. Continuous signed colour uses a symmetric asinh($G$) mapping, with $G$ in percentage points, as a nonlinear visual aid only. No values are clipped; the printed raw percentages at the reported precision are authoritative. Legend ticks are raw percentages. Colour implies neither significance nor a consistency class. Materna T1, T2 and T3 are dependent temporal blocks. Comparisons are bounded to the evaluated implementation and eligible populations.

and disk write conditionally positive; CPU and throughput were neutral or mixed. The memory bootstrap interval excluded zero, whereas the near-zero throughput interval crossed it. In Rnd, CPU and memory were conditionally positive, while throughput and disk write were neutral or mixed. All four Rnd intervals crossed zero, including the interval for its negative throughput point estimate.

The Materna blocks further changed the resource pattern. T1 memory was approximately neutral and slightly negative (−0.02%); only throughput reached the conditionally positive class. The T1 CPU interval was entirely negative, while the other three intervals crossed zero. T2 CPU and memory were broadly positive with intervals above zero; throughput and disk write remained neutral or mixed. T3 memory and disk write were conditionally positive, but all four T3 intervals crossed zero.

Memory has the clearest recurring positive pattern; the strongest benefit is not always attached to one resource. Rnd throughput and T1 disk write illustrate adverse effects where other blocks are near zero or positive.

*Post-hoc cross-model and seed checks.* The matched-context Ridge ARX:AR comparison (Table 3B) also shows selective contextual value, but individual directions depend on model family and environment. Memory improves under Ridge in all five external blocks; fastStorage, Rnd and Materna T2/T3 have positive point gains in both model families. TimeTrack memory instead reverses under Ridge, and Materna T1's near-zero negative LSTM estimate contrasts with a positive Ridge gain. Memory S2:S1 gains in fastStorage, Rnd and Materna T2 and T3 are positive in all three recorded seeds (Supplementary Table S2). External memory is thus the strongest recurring favourable contextual pattern, not a universal benefit. Existing-seed diagnostics find sign changes in 9/24 LSTM S2:S1 block/resource summaries across the three fixed seeds (Supplementary Table S2). Point gains or reversals with seed-varying signs should not be treated as seed-invariant improvements or temporal regime changes.

The Ridge TimeTrack throughput gain of −838.45% is diagnosed as CONCENTRATED DEGRADATION, not broad Ridge failure. The saved-artifact audit attributes 99.93% of positive excess error to machine06/fold3. ARX nevertheless improves 21/35 machine/fold pairs, with median ARX/AR MAE ratio approximately 0.9963. Predictions and primary MAEs are finite, and transforms, scalers and support match the frozen design; no implementation defect is demonstrated.

The aggregate penalty must be read with this concentration diagnosis rather than as a typical-machine effect.

## 5.2. RQ2 Operational Shared-Representation Comparison

RQ2 compares M1 with S2, holding the all-resource input scope fixed and changing whether the predictive representation is shared. Throughput, memory and disk write had negative primary gains in all six blocks (Figure 2(b)). CPU was positive only in TimeTrack, fastStorage and Materna T1. These small or near-zero exceptions do not support uniformly beneficial sharing within the evaluated M1 design.

In TimeTrack, memory was broadly negative; the remaining frozen resource classifications were neutral or mixed. In fastStorage, CPU was neutral or mixed and its bootstrap interval crossed zero. Throughput, memory and disk write were broadly negative with intervals below zero. Rnd was broadly negative for every resource, and each interval also lay below zero.

Materna T1 CPU was neutral or mixed with an interval crossing zero, whereas the other resources were broadly negative with intervals wholly below zero. T2 was broadly negative for all four resources, with entirely negative overall intervals. In T3, throughput, memory and disk write were broadly negative; CPU was neutral or mixed under the frozen consistency rule despite its negative primary gain. All four T3 intervals excluded zero, with the disk-write interval spanning $[-239.63\%, -5.94\%]$.

Degradation recurred outside the CPU exceptions, with resource- and block-dependent magnitudes. Point signs, consistency classes and intervals remain distinct under the frozen protocol.

*Post-hoc sharing controls and matched cells.* Table 3A shows H64 M1 against S2 constrained to the frozen H64-M1 contexts, and H129 M1 against original S2. Throughput, memory and disk-write point estimates remain adverse in all six blocks under both separate controls. Context-selection freedom and the original aggregate capacity difference are therefore incomplete explanations within this pipeline. The controls do not identify sharing alone as the cause: architecture, optimisation and training objectives remain part of the implemented comparison, and common-S2 contexts align H64, not necessarily H129. Existing-seed diagnostics find sign changes in 4/24 original M1:S2 block/resource summaries (Supplementary Table S2), concentrated in small CPU exceptions and TimeTrack throughput. Adverse aggregate directions do not imply adverse outcomes for every seed. Here, "stable" means "no sign flip across the three recorded seeds". Sign stability does not imply magnitude stability: Materna T3 M1:S2 disk-write gains are $-54.65\%$, $-71.55\%$ and $-87.84\%$ (Supplementary Table S2).

The post-hoc matched-cell diagnostics further separate direction from magnitude. A cell is entity $\times$ outer fold $\times$ forecast lead, with MAE first averaged across existing seeds. In all 24 original M1:S2 block/resource summaries (Supplementary Table S1), the median candidate/reference MAE ratio exceeds one and the candidate win rate is below 50%. This includes the small positive CPU MacroMAE exceptions. Materna T1/T3 disk-write median ratios are approximately $1.0246/1.0347$, despite MacroMAE gains of $-27.46\%/-71.30\%$. The adverse typical-cell direction is widespread, but the much larger mean-error penalties do not describe every VM or the magnitude of degradation in a typical cell.

Post-hoc attribution locates this uneven magnitude. In Materna T1/T3, 28/35 and 32/35 VM-averaged disk-write errors worsen. VM266 dominates the T1 tail and VM272 the T3 tail, contributing 82.10% and 86.97% of positive excess error, respectively; the top three VMs contribute 91.72% and 92.93%. No training or scaling defect was demonstrated. A post-hoc frozen-partition audit, using each VM's frozen M1-selected context within each fold, places their log-space training IQRs near the upper cohort tail: VM266 ranks 33–34/35 at approximately 7.86–8.19 times the cohort median IQR, and VM272 ranks 35/35 at 7.59–7.75 times it. Both have the largest native disk-write IQR in every fold. Unusually small training IQRs are therefore not supported as an explanation for these tails. This does not establish that the large scale caused the errors or that sharing is causal. For TimeTrack memory, all seven machine averages and 31/35 machine/fold pairs worsen, with median pair ratio 1.195276. The largest pair contributes 21.80% and the top three 51.06% of positive excess error at that same machine/fold level. Adverse direction is broad but magnitude uneven; pair-level attribution must not be conflated with the finer matched-cell medians in Supplementary Table S1.

## 5.3. RQ3 Full Joint-System Performance

RQ3 compares the complete M1 system with local-resource S1 models. Figure 2(c) is negative in every displayed cell except Materna T2 CPU, so the full joint formulation generally did not outperform S1. In TimeTrack, memory and disk write were broadly negative; CPU and throughput were neutral or mixed under the prospective consistency criteria.

In fastStorage, CPU and throughput were broadly negative, whereas memory and disk write were neutral or mixed. The throughput interval was entirely below zero, while the other three overall intervals crossed zero. In Rnd, every resource was broadly negative and every interval lay below zero.

In Materna T1, throughput and disk write were broadly negative; CPU and memory were neutral or mixed, although all four VM-cluster intervals lay below zero. The T2 CPU exception ($+1.75\%$) was conditionally positive, but its interval crossed zero ($[-3.52\%, +6.07\%]$). T2 disk write was broadly negative, and throughput and memory were neutral or mixed. In T3, CPU, throughput and disk write were broadly negative, while memory was neutral or mixed; only the memory interval narrowly crossed zero. These differences in classification and uncertainty remain important despite the nearly uniform negative sign pattern.

**Table 3**
POST-HOC sensitivity comparisons alongside frozen reference gains (%).

| **A:** [H64 M1:S2 / H64 M1:common-S2 / H129 M1:S2] | | | | |
|---|---|---|---|---|
| Block | CPU | Throughput | Memory | Disk write |
| TimeTrack | [+0.47/ − 0.81/ − 2.93] | [−0.84/ − 0.54/ − 0.16] | [−18.48/ − 17.79/ − 18.48] | [−0.80/ − 0.56/ − 0.60] |
| fastStorage | [+0.05/ + 0.36/ + 3.14] | [−27.09/ − 20.31/ − 33.83] | [−4.87/ − 4.43/ − 4.19] | [−4.07/ − 1.75/ − 4.05] |
| Rnd | [−16.51/ − 16.32/ − 17.57] | [−14.32/ − 12.57/ − 15.56] | [−8.92/ − 8.69/ − 8.06] | [−5.55/ − 3.72/ − 6.85] |
| Materna T1 | [+0.88/ − 0.61/ + 3.19] | [−7.31/ − 7.19/ − 4.72] | [−13.22/ − 10.15/ − 12.32] | [−27.46/ − 27.98/ − 24.62] |
| Materna T2 | [−2.56/ − 1.38/ + 0.17] | [−8.69/ − 7.26/ − 3.65] | [−9.27/ − 7.52/ − 6.72] | [−7.89/ − 7.93/ − 7.20] |
| Materna T3 | [−8.10/ − 7.40/ − 8.21] | [−7.91/ − 7.44/ − 5.83] | [−8.70/ − 7.79/ − 6.53] | [−71.30/ − 69.67/ − 64.00] |
| **B:** [frozen LSTM S2:S1 / Ridge ARX:AR] | | | | |
| Block | CPU | Throughput | Memory | Disk write |
| TimeTrack | [−3.31/ − 14.00] | [−0.24/ − 838.45][a] | [+0.86/ − 23.21] | [−0.49/ − 7.58] |
| fastStorage | [−3.20/ − 4.37] | [+0.01/ − 20.75] | [+2.45/ + 2.88] | [+2.69/ + 1.05] |
| Rnd | [+1.41/ + 0.22] | [−7.21/ + 0.37] | [+1.67/ + 4.26] | [+0.16/ + 1.68] |
| Materna T1 | [−4.32/ + 6.27] | [+0.99/ − 13.90] | [−0.02/ + 7.22] | [−6.70/ + 2.05] |
| Materna T2 | [+4.21/ + 9.67] | [+0.29/ + 0.91] | [+3.95/ + 6.47] | [−1.12/ + 10.07] |
| Materna T3 | [−2.79/ + 2.17] | [+0.04/ − 2.15] | [+3.09/ + 1.81] | [+1.91/ − 0.69] |

Positive means lower candidate MacroMAE; values are rounded to two decimals. Frozen H64 M1:S2 and LSTM S2:S1 references are repeated unchanged, not replaced by controls. For TimeTrack memory, both H64 and H129 gains round to −18.48%; they are distinct before rounding. Common-S2 uses H64-M1 contexts; H129 selects its own contexts. These are separate controls, not joint causal isolation. [a]The TimeTrack Ridge throughput penalty is concentrated in machine06/fold3: 99.93% of positive excess error; ARX improves 21/35 machine/fold pairs, median ARX/AR ratio 0.9963 (Section 5.1). Point signs imply neither significance nor model-independent benefit; Materna T1–T3 remain dependent blocks.

RQ3 combines both design changes. Its gains are reported directly, not added from RQ1 and RQ2 percentages (Section 3.6).

## 5.4. RQ4 Resource, Environment, and Temporal Heterogeneity

Figure 2 reveals heterogeneity at each prespecified level. Resource heterogeneity is evident in the selective S2-versus-S1 benefits and changing resource ordering. CPU, throughput and disk write change direction across evidence blocks, whereas memory is positive in five and nearly zero and negative only in Materna T1. Repeated positive memory effects therefore do not imply invariant resource behaviour.

External-environment heterogeneity is stronger for representation sharing: fastStorage CPU was near zero while the other three resources were negative; Rnd was broadly negative for all four. Temporal heterogeneity within the dependent Materna blocks is distinct: M1-versus-S2 CPU changed from positive in T1 to negative in T2–T3, moving from neutral or mixed to broadly negative and back to neutral or mixed. Throughput, memory and disk write were broadly negative in all three Materna blocks, with varying magnitudes. For M1 versus S1, CPU alone reversed sign in T2; throughput and disk write stayed negative and memory stayed negative but neutral or mixed across T1–T3.

Sign, magnitude, and frozen class capture different aspects of this heterogeneity. For example, T3 S2-versus-S1 throughput was slightly positive at +0.04% but remained neutral or mixed, while T2 CPU was both positive and broadly positive. T3 M1-versus-S2 CPU had a negative point estimate and an interval below zero, yet remained neutral or mixed under the separate consistency rule. These combinations prevent near-zero sign changes from being treated as equivalent to changes supported across the prescribed units. They also show why the within-Materna decision uses the frozen classification and trace rules alongside the raw gains.

The frozen within-Materna RQ4 outcome was recorded as follows.

The prespecified RQ4 criterion for temporal heterogeneity was met. Seven of the 12 resource–comparison pairs changed frozen classification across T1–T3 and five changed raw sign. The S2-versus-S1 trace decision also changed: T1 met the frozen non-support rule, whereas T2 and T3 were mixed. The M1-versus-S2 trace-level decision was consistently not supported, even though its CPU classification varied. Several small S2 effects are seed-variable (Supplementary Table S2). This is a descriptive criterion, not an IID or inferential claim; no post-hoc temporal regimes or causal mechanisms are inferred.

## 5.5. Persistence Boundary

Relative formulation gains do not establish superiority to a persistence forecast. In TimeTrack, S1, S2, and M1 all had positive mean persistence skill for throughput and disk write, and their CPU means were also positive. Memory was the boundary: mean skill was −0.1272 for S1, −0.1494 for S2, and −0.6047 for M1, with positive skill at only two, four, and zero of eight leads. Thus the large negative M1-versus-S2 memory result coincided with weak absolute performance

**Table 4**
Local model complements and summed Materna worker fit-hours.

| Formulation | Models | Parameters | Worker h |
|---|---|---|---|
| S1 | 4 | 203,028 | 85.59 |
| S2 | 4 | 206,100 | 75.30 |
| M1 | 1 | 52,500 | 20.41 |

Models and parameters describe one entity/fold/seed target complement in the five-horizon external design. Worker hours sum successful inner/final fits across all three Materna traces, not elapsed wall time or deployment cost. Relative to S2, H64 M1 uses approximately 74.5% fewer parameters and 75% fewer models.

relative to persistence, whereas the throughput and disk-write formulation comparisons occurred in resources with positive average absolute skill.

Neither Bitbrains environment contained a formulation that beat persistence for every resource. Overall CPU skill was negative for S1, S2, and M1 in both environments. In fastStorage, throughput skill was also negative for every formulation; memory and disk write were positive for all three. Rnd again had positive memory and disk-write skill for all formulations, while throughput was only slightly positive for S1 (0.0384) and negative for S2 (−0.0257) and M1 (−0.1176). Consequently, a positive S2-versus-S1 gain for Rnd CPU or a relative degradation under M1 does not by itself describe absolute forecasting skill.

Materna showed a different aggregate boundary. S2 beat persistence in all 12 trace–resource overall aggregates. No formulation, however, beat persistence in every individual trace–resource–lead cell. At the overall level, M1 failed the check for T1 throughput (−0.0608), T1 disk write (−1.7590), T3 throughput (−0.2629), and T3 disk write (−1.3493); it remained positive for all four T2 resources. S1 additionally failed only T1 throughput at the overall level, while S2 had no overall failures. These persistence results delimit absolute forecasting performance and remain separate from the controlled S1–S2–M1 comparisons.

Thus relative gain, overall persistence skill and horizon-level skill answer different questions (Section 4.8); none establishes uniform absolute forecasting superiority.

### 5.6. Accuracy-Efficiency Trade-Off

The original H64 M1 consolidates the target-model complement for one entity, outer fold and seed (Table 4). In the five-horizon external design, S1 uses four networks and 203,028 parameters, S2 four networks and 206,100 parameters, and M1 one joint network and 52,500 parameters: approximately 74.5% fewer parameters and 75% fewer models than S2. TimeTrack's eight-horizon complements contain 206,880 S2 parameters and 53,280 M1 parameters. These are local architecture inventories, not a globally deployed four-versus-one system or total training-fit counts. The post-hoc H129 comparison changes the parameter budget and must not inherit H64's parameter-reduction claim.

The frozen Materna totals are 85.594240 worker fit-hours for S1, 75.303186 for S2 and 20.405083 for M1. As defined in Section 4.6, these sum successful inner-candidate and final-refit durations, including inner validation, across the three traces. They are not elapsed wall time, energy, deployment cost or production latency. Different executed epochs and selected contexts can contribute to the S1–S2 order; input width alone does not explain it. The measured training workload and local model consolidation coexist with the accuracy penalties above, without establishing production savings or an architecture-independent efficiency ranking.

## 6. Cross-Dataset Synthesis

Table 5 retains frozen TimeTrack, equal-environment Bitbrains and equal-temporal-block Materna summaries under Section 4.9, without a global score. The post-hoc evidence in Sections 5.1–5.2 qualifies their interpretation; agreement concerns operational directions, not independent replication of a sharing mechanism.

### 6.1. TimeTrack

TimeTrack's small positive S2:S1 memory and M1:S2 CPU estimates vary across seeds. Ridge reverses the contextual memory direction; its throughput penalty is concentrated in one machine/fold (Section 5.1). This is development evidence, not independent external confirmation.

### 6.2. Bitbrains External Confirmation

Memory has favourable LSTM/Ridge contextual point agreement in both external environments, whereas throughput reverses direction between families in each. Their equal-environment summary must not erase these block-specific differences.

### 6.3. Materna Temporal Replication

LSTM/Ridge memory gains agree favourably in T2/T3, but T1's near-zero, seed-variable LSTM estimate contrasts with positive Ridge gain. Other resource directions depend on block and family. These consecutive blocks retain the same selected VM identifiers; their dependence and the T1/T3 disk-write magnitude qualifications (Section 5.2) preclude independent replication or uniformly severe VM-degradation claims.

### 6.4. Generalisation Boundaries

*Contextual inputs.* External memory recurs favourably, but cross-family reversals and seed-variable small effects require resource-, environment- and family-specific validation.

*Shared representation.* Both separate controls retain adverse throughput, memory and disk-write points. Median MAE ratios and win rates retain the adverse typical-cell direction despite small positive CPU MacroMAE gains. Capacity and selection are incomplete explanations, not eliminated causal factors (Section 5.2).

*Complete joint system and consolidation.* M1:S1 gains are negative except Materna T2 CPU. Local model counts

**Table 5**
Frozen descriptive source summaries of primary overall MacroMAE gain (%).

| | S2 versus S1 | | | M1 versus S2 | | | M1 versus S1 | | |
|---|---|---|---|---|---|---|---|---|---|
| Resource | TimeTrack | Bitbrains | Materna | TimeTrack | Bitbrains | Materna | TimeTrack | Bitbrains | Materna |
| CPU | −3.31 | −0.89 | −0.97 | 0.47 | −8.23 | −3.26 | −2.83 | −9.01 | −4.25 |
| Throughput | −0.24 | −3.60 | 0.44 | −0.84 | −20.71 | −7.97 | −1.08 | −24.82 | −7.50 |
| Memory | 0.86 | 2.06 | 2.34 | −18.48 | −6.89 | −10.40 | −17.46 | −4.70 | −7.85 |
| Disk write | −0.49 | 1.42 | −1.97 | −0.80 | −4.81 | −35.55 | −1.29 | −3.33 | −37.71 |

Positive values indicate lower MacroMAE for the candidate formulation; negative values indicate higher MacroMAE. TimeTrack: development summary. Bitbrains: equal-environment synthesis of fastStorage and Rnd. Materna: equal-temporal-block synthesis of T1–T3, consecutive dependent temporal blocks rather than independent replications. Comparisons are descriptive, without cross-source aggregation.

and worker fit-hours describe a consolidation/workload trade-off, not production benefit (Section 5.6). Population, model-family, seed and persistence limits are assessed in Section 8; the controls neither change the endpoints nor extend evidence to excluded VMs or untested shared architectures.

# 7. Discussion

## 7.1. When Cross-Resource Context Helps

Cross-family agreement for external memory, TimeTrack's reversal and T1's seed-variable estimate support selective contextual value, not a universal memory benefit (Section 5.1). Small seed-variable effects cannot establish stable improvements or temporal regimes.

This agrees with MMRP's non-uniform contextual gains (Tripathi et al., 2025) and the relevance of environment-specific evaluation (Rossi et al., 2025). TimeTrack correlations (Meliani et al., 2025) motivate inputs, not conditional predictive value or causal resource coupling.

The concentrated Ridge throughput penalty in Section 5.1 shows why a large aggregate loss must not be generalised to typical-machine failure; the saved-artifact checks demonstrated no implementation defect.

## 7.2. Shared Representation and Adverse Transfer

The separate sensitivity controls retain adverse throughput, memory and disk-write points across all six blocks (Section 5.2). Context selection and aggregate capacity contribute to magnitudes but do not suffice to explain the pattern.

Direction need not imply uniform severity. Materna T1/T3 disk-write median ratios are about 1.025/1.035, despite much larger tail-sensitive MacroMAE penalties. Direction is broadly adverse but absolute-error magnitude concentrated, not equally severe in every VM (Section 5.2).

The outcome is operationally consistent with adverse transfer, not a uniquely identified sharing mechanism. Useful task-dependent transfer is established (Caruana, 1997; Standley et al., 2020; Fifty et al., 2021), including in forecasting (Nikentari and Wei, 2022; Nochumsohn et al., 2025). Gradient conflict, representation interference and capacity competition were not measured. Nominal loss weights do not establish realised contributions (Section 3.5); architecture, optimisation and selection remain coupled. Multi-objective or gradient-aware designs (Sener and Koltun, 2018; Yu et al., 2020) are alternatives, not diagnoses or demonstrated remedies for these models.

## 7.3. Implications for Runtime and Cloud Software Systems

Forecasting supports adaptation (Taherizadeh et al., 2018; Bauer et al., 2020), but adding telemetry and consolidating predictors require separate resource- and horizon-specific validation on future-only support, including persistence (Section 5.5). A contextual benefit does not imply a sharing benefit. Controller outcomes depend on policy and platform (Qu et al., 2018); no autoscaler, scheduler or controller was deployed. Production latency, energy, monetary savings, SLA improvements and controller benefits were not measured.

## 7.4. Accuracy-Consolidation Trade-Off

The local H64 model/parameter reductions and lower Materna worker fit-hours coexist with accuracy losses (Section 5.6). H129 retains consolidation but not H64's parameter reduction. Compactness is therefore a trade-off, not a free predictive gain. Joint accuracy/computation assessment follows task-grouping and service-forecasting research (Standley et al., 2020; Syu et al., 2017); deployment still requires platform measurements and resource-specific error tolerances.

Per-entity consolidation may still be attractive where modest target-specific accuracy penalties are acceptable. This requires distinguishing typical-cell magnitude from tail-sensitive MacroMAE penalties and checking adverse tails, not recommending M1 universally or treating concentrated excess as irrelevant.

## 7.5. Applicability and Generalisation Limits

The population and dependence limits in Sections 4.10 and 8.3 bound applicability. Cross-family reversals do not select a universal replacement. Selective grouping and adaptive or hybrid sharing remain untested alternatives (Fifty et al., 2021; Yu et al., 2020; Nochumsohn et al., 2025). Global models remain outside this evaluation (Montero-Manso and Hyndman, 2021). The same limit applies to foundation forecasters (Di Menna et al., 2026). Future comparisons should retain independent all-input controls

and resource-specific assessment while measuring proposed mechanisms.

## 8. Threats to Validity

### 8.1. Construct Validity

The three formulations are not a complete $2 \times 2$ design. Original capacity/context differences are addressed separately, not jointly, by the post-hoc controls (Section 4.11). H129's fresh selection and H64-aligned common-S2 contexts leave residual differences; parameter-count proximity is not architectural, task-capacity or optimisation equivalence.

CPU, throughput, memory and disk-write mappings have source-specific definitions and native units, not perfect measurement equivalence. MacroMAE gains describe the specified error estimand; RMSE, $R^2$, native NMAE and persistence provide complementary diagnostics. Metric choice can alter rankings (Hyndman and Koehler, 2006). Scaled-space training and selection objectives are distinct from native-unit evaluation (Sections 4.7–4.8), and a relative formulation gain need not establish useful absolute skill.

### 8.2. Internal Validity

The chronology, purging and training-only preprocessing in Section 4.4 constrain leakage (Cerqueira, Torgo and Mozetič, 2020; Hyndman and Athanasopoulos, 2021). Matched support and provenance cannot exclude all implementation/data errors. Original external criteria were prospective; reviewer-motivated controls are post-hoc, not prespecified confirmation.

Only three neural seeds were recorded. Existing diagnostics identify 9/24 S2:S1 and 4/24 M1:S2 block/resource summaries whose gains change sign across those seeds (Supplementary Table S2). These counts do not establish a general noise floor or stability under arbitrary retraining. Contexts, epochs and optimisation can interact with formulation, and missing gradient/clipping histories limit mechanism attribution even after the separate controls.

Ridge is deterministic but restricted to a finite five-alpha grid and frozen H64-M1 contexts, which were selected for the LSTM design rather than optimised for Ridge. Boundary alpha selections do not establish performance under a wider regularisation search. Its selection uses native-unit validation MAE, unlike the LSTM context score, so cross-family disagreement is conditional on these implemented designs, not an architecture-only intervention.

### 8.3. External Validity

The sources represent particular runtime/cloud contexts, not probability samples of all VMs. Excluded workloads may differ (Section 4.10). Materna's shared source and VM identifiers make temporal agreement repeated-within-source evidence, not independent-dataset replication.

The evaluated model families comprise the conventional LSTM formulations, their width/context controls and the matched-context Ridge pair. Reversals across these designs preclude model-independent contextual claims. Transformers, mixture-of-experts, selective grouping, adaptive sharing, gradient surgery and foundation forecasters were not tested. Four resource semantics do not establish accuracy for application latency, queue lengths, QoS or downstream SLA outcomes.

### 8.4. Conclusion Validity

MacroMAE weights seed-averaged entity–fold cells and horizons equally, but large native errors affect it differently from median ratios and win rates. The direction/magnitude distinction in Section 5.2 precludes uniform VM-level interpretations; diagnostics do not replace primary estimates.

Paired VM-cluster bootstrap intervals condition on the evaluated external cohorts and protocol. They are separate from frozen consistency classes and point signs, not classical significance claims or intervals retrospectively assigned to every post-hoc control. Overlapping expanding folds, multiple horizons, repeated seeds and dependent Materna blocks are not IID replicates. No six-block global effect, independent-dataset meta-analysis or IID interpretation is used to strengthen these conclusions.

Persistence failures in Section 5.5 limit absolute skill claims. Neither relative gains nor descriptive source summaries support a universal verdict beyond the evaluated formulations and populations.

## 9. Conclusion

Contextual input and representation sharing require separate design decisions. External memory recurs favourably, but Ridge and seed sensitivity bound contextual value by model and environment. The evaluated shared-LSTM model generally loses throughput, memory and disk-write accuracy while consolidating per-entity models. Adverse point directions survive separate post-hoc common-context and H129 approximate-capacity controls, without identifying sharing alone as the mechanism. Typical-cell penalties can be more modest than tail-sensitive MacroMAE suggests: per-entity consolidation may be attractive when target-specific accuracy penalties are acceptable, provided adverse tails are assessed. These findings do not universally reject multitask forecasting. Validate contextual input and consolidation by resource and design, within the eligible population and against absolute persistence skill.

## Data availability

TimeTrack is publicly available on Kaggle (Apache 2.0 in its dataset metadata) and in the publisher's Zenodo archive (doi:10.5281/zenodo.15442251; CC BY 4.0); its description paper is Meliani et al. (2025). These are source-specific licence statements. Bitbrains fastStorage and Rnd are publicly available as GWA-T-12 (Shen, van Beek and Iosup, 2015); Materna T1–T3 are publicly available as GWA-T-13 (Kohne, Spohr, Nagel and Spinczyk, 2014; Kohne, Pasternak, Nagel and Spinczyk, 2016). We acknowledge Bitbrains

IT Services Inc. and Materna GmbH Information & Communications, respectively, for providing these traces through the Grid Workloads Archive. Their separate source pages specify acknowledgement and citation conditions, rather than a standard open-data licence. Readers should obtain third-party raw data from these original sources and consult each source's terms before redistribution.

Source code, protocols, derived analysis outputs and SHA-256 inventories are publicly accessible in the project repository at scientific checkpoint `fee20c3`. The public checkpoint does not include the full fitted-model or per-fit prediction collections; the checksum inventories do not themselves provide those files. No repository-wide reuse licence is asserted here.

## CRediT authorship contribution statement

**Abdelhak Kadouma:** Conceptualization, Methodology, Software, Data curation, Formal analysis, Validation, Visualization, Writing – original draft. **Amit Shukla:** Supervision, Conceptualization, Methodology, Writing – review & editing. **Mohammed Elmusrati:** Supervision, Project administration, Writing – review & editing.

## Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

## References

Bauer, A., Züfle, M., Herbst, N., Zehe, A., Hotho, A., Kounev, S., 2020. Time series forecasting for self-aware systems. Proceedings of the IEEE 108, 1068–1093. doi:10.1109/JPROC.2020.2983857.

Bi, J., Yuan, H., Zhou, M., Liu, Q., 2019. Time-dependent cloud workload forecasting via multi-task learning. IEEE Robotics and Automation Letters 4, 2401–2406. doi:10.1109/LRA.2019.2899224.

Calheiros, R.N., Masoumi, E., Ranjan, R., Buyya, R., 2015. Workload prediction using ARIMA model and its impact on cloud applications' QoS. IEEE Transactions on Cloud Computing 3, 449–458. doi:10.1109/TCC.2014.2350475.

Caruana, R., 1997. Multitask learning. Machine Learning 28, 41–75. doi:10.1023/A:1007379606734.

Cerqueira, V., Torgo, L., Mozetič, I., 2020. Evaluating time series forecasting models: an empirical study on performance estimation methods. Machine Learning 109, 1997–2028. doi:10.1007/s10994-020-05910-7.

Di Menna, F., Traini, L., Cortellessa, V., 2026. Forecasting software runtime metrics: A comparative study of classical statistical, neural network, and foundation models. Journal of Systems and Software 240, 112937. doi:10.1016/j.jss.2026.112937.

Fifty, C., Amid, E., Zhao, Z., Yu, T., Anil, R., Finn, C., 2021. Efficiently identifying task groupings for multi-task learning, in: Advances in Neural Information Processing Systems, Curran Associates, Inc.. pp. 27503–27516. URL: https://proceedings.neurips.cc/paper/2021/hash/e77910ebb93b511588557806310f78f1-Abstract.html.

Gao, M., Yu, Z., Luo, Y., Ma, L., Sun, Y., Fang, Y., Gao, X., 2026a. Macro-micro collaborative learning for logical data center microservice indicators forecasting, in: Proceedings of the ACM Web Conference 2026, ACM. pp. 5010–5020. doi:10.1145/3774904.3792125.

Gao, X., Tang, X., Yao, C., Wu, S., Yuan, G., Zhou, W., Li, F., Chen, G., 2026b. OMCR: An online multivariate forecaster for cloud resource management. IEEE Transactions on Knowledge and Data Engineering 38, 532–545. doi:10.1109/TKDE.2025.3619097.

Hochreiter, S., Schmidhuber, J., 1997. Long short-term memory. Neural Computation 9, 1735–1780. doi:10.1162/neco.1997.9.8.1735.

Hyndman, R.J., Athanasopoulos, G., 2021. Forecasting: Principles and Practice. 3rd ed., OTexts, Melbourne, Australia. URL: https://otexts.com/fpp3/. accessed 2026-10-16.

Hyndman, R.J., Koehler, A.B., 2006. Another look at measures of forecast accuracy. International Journal of Forecasting 22, 679–688. doi:10.1016/j.ijforecast.2006.03.001.

Kohne, A., Pasternak, D., Nagel, L., Spinczyk, O., 2016. Evaluation of SLA-based decision strategies for VM scheduling in cloud data centers, in: Proceedings of the 3rd Workshop on CrossCloud Infrastructures & Platforms, ACM. pp. 6:1–6:5. doi:10.1145/2904111.2904113.

Kohne, A., Spohr, M., Nagel, L., Spinczyk, O., 2014. FederatedCloudSim: A SLA-aware federated cloud simulation framework, in: Proceedings of the 2nd International Workshop on CrossCloud Systems, ACM. pp. 3:1–3:5. doi:10.1145/2676662.2676674.

Liu, Y., Hu, T., Zhang, H., Wu, H., Wang, S., Ma, L., Long, M., 2024. iTransformer: Inverted transformers are effective for time series forecasting, in: The Twelfth International Conference on Learning Representations. URL: https://openreview.net/forum?id=JePfAI8fah.

Meliani, A.E., Arora, S., Ksentini, A., Knopp, R., 2025. TimeTrack: A dataset for exploring temporal patterns and predictive insights into OpenAirInterface (OAI) CI/CD cluster, in: ICC 2025 - IEEE International Conference on Communications, IEEE. pp. 1978–1983. doi:10.1109/ICC52391.2025.11161298.

Montero-Manso, P., Hyndman, R.J., 2021. Principles and algorithms for forecasting groups of time series: Locality and globality. International Journal of Forecasting 37, 1632–1653. doi:10.1016/j.ijforecast.2021.03.004.

Nie, Y., Nguyen, N.H., Sinthong, P., Kalagnanam, J., 2023. A time series is worth 64 words: Long-term forecasting with transformers, in: The Eleventh International Conference on Learning Representations. URL: https://openreview.net/forum?id=Jbdc0vTOcol.

Nikentari, N., Wei, H.L., 2022. Multi-task learning for time series forecasting using NARMAX-LSTM, in: 2022 27th International Conference on Automation and Computing (ICAC), IEEE. pp. 1–6. doi:10.1109/ICAC55051.2022.9911071.

Nochumsohn, L., Zisling, H., Azencot, O., 2025. A multi-task learning approach to linear multivariate forecasting, in: Proceedings of The 28th International Conference on Artificial Intelligence and Statistics, PMLR. pp. 2638–2646. URL: https://proceedings.mlr.press/v258/nochumsohn25a.html.

Qu, C., Calheiros, R.N., Buyya, R., 2018. Auto-scaling web applications in clouds: A taxonomy and survey. ACM Computing Surveys 51, 73:1–73:33. doi:10.1145/3148149.

Rossi, A., Visentin, A., Carraro, D., Prestwich, S., Brown, K.N., 2025. Forecasting workload in cloud computing: towards uncertainty-aware predictions and transfer learning. Cluster Computing 28, 258. doi:10.1007/s10586-024-04933-2.

Salinas, D., Flunkert, V., Gasthaus, J., Januschowski, T., 2020. DeepAR: Probabilistic forecasting with autoregressive recurrent networks. International Journal of Forecasting 36, 1181–1191. doi:10.1016/j.ijforecast.2019.07.001.

Sener, O., Koltun, V., 2018. Multi-task learning as multi-objective optimization, in: Advances in Neural Information Processing Systems, Curran Associates, Inc.. pp. 525–536. URL: https://proceedings.neurips.cc/paper/2018/hash/432aca3a1e345e339f35a30c8f65edce-Abstract.html.

Shen, S., van Beek, V., Iosup, A., 2015. Statistical characterization of business-critical workloads hosted in cloud datacenters, in: 2015 15th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing, IEEE. pp. 465–474. doi:10.1109/CCGrid.2015.60.

Standley, T., Zamir, A., Chen, D., Guibas, L., Malik, J., Savarese, S., 2020. Which tasks should be learned together in multi-task learning?, in: Proceedings of the 37th International Conference on Machine Learning, PMLR. pp. 9120–9132. URL: https://proceedings.mlr.press/v119/standley20a.html.

Syu, Y., Kuo, J.Y., Fanjiang, Y.Y., 2017. Time series forecasting for dynamic quality of web services: An empirical study. Journal of Systems

and Software 134, 279–303. doi:10.1016/j.jss.2017.09.011.

Taherizadeh, S., Jones, A.C., Taylor, I., Zhao, Z., Stankovski, V., 2018. Monitoring self-adaptive applications within edge computing frameworks: A state-of-the-art review. Journal of Systems and Software 136, 19–38. doi:10.1016/j.jss.2017.10.033.

Tran, N., Nguyen, T., Nguyen, B.M., Nguyen, G., 2018. A multivariate fuzzy time series resource forecast model for clouds using LSTM and data correlation analysis. Procedia Computer Science 126, 636–645. doi:10.1016/j.procs.2018.07.298.

Tripathi, S., Priyadarshni, Misra, R., Singh, T.N., 2025. Multilayer multivariate forecasting network for precise resource utilization prediction in edge data centers. Future Generation Computer Systems 166, 107692. doi:10.1016/j.future.2024.107692.

Ullah, F., Bilal, M., Yoon, S.K., 2023. Intelligent time-series forecasting framework for non-linear dynamic workload and resource prediction in cloud. Computer Networks 225, 109653. doi:10.1016/j.comnet.2023.109653.

Xu, M., Song, C., Wu, H., Gill, S.S., Ye, K., Xu, C., 2022. esDNN: Deep neural network based multivariate workload prediction in cloud computing environments. ACM Transactions on Internet Technology 22, 75:1–75:24. doi:10.1145/3524114.

Yu, J., Liu, X., Luo, C., Huang, J., Zhou, R., Liu, Y., Hu, J., Chen, J., Zhang, K., Zhang, D., Shen, Y., Adhikarla, E., Dai, Y., Zhang, K., Kong, Z., Ye, W., Yin, Y., Namboodiri, V., Davison, B., Moore, J., Chen, Y., 2025. Multitask learning 1997–2024: Part I fundamentals. Harvard Data Science Review 7. URL: https://hdsr.mitpress.mit.edu/pub/7fcc3jhv/release/2, doi:10.1162/99608f92.462f48a2.

Yu, T., Kumar, S., Gupta, A., Levine, S., Hausman, K., Finn, C., 2020. Gradient surgery for multi-task learning, in: Advances in Neural Information Processing Systems, Curran Associates, Inc.. pp. 5824–5836. URL: https://proceedings.neurips.cc/paper/2020/hash/3fe78a8acf5fda99de95303940a2420c-Abstract.html.

Zhang, B., Yang, G., Yu, F., Pan, Q., 2025. SWT-CLSTM: A hybrid model for cloud workload prediction combining smooth wavelet transform and contrastive learning. Journal of King Saud University Computer and Information Sciences 37, 320. doi:10.1007/s44443-025-00316-8.